Finance Project | Insurance Claim Fraud Detection & Severity Prediction

Domain: InsurTech / General Insurance | Difficulty: Hard | Industry Demand: High

Problem Statement

Insurance fraud costs the Indian insurance industry an estimated ₹30,000 crore annually (15–20% of total premiums). Current fraud investigation is manual and reactive — claims are investigated only after payment, recovering <30% of fraudulent payouts. The challenge is to build a proactive ML system that (1) flags claims likely to be fraudulent before payment and (2) predicts the expected claim settlement amount for reserve setting — enabling insurers to prioritise Special Investigation Unit (SIU) resources on highest-risk claims.

Project Overview

Two-model architecture: (1) classify fraudulent claims before payment and (2) predict claim payout amount for reserve setting. Insurance companies (LIC, HDFC Ergo, Star Health, Bajaj Allianz) and reinsurers (Munich Re, Swiss Re India) are actively building this. The project uses both classification and regression pipelines with shared feature engineering.

Why This Is Challenging

Soft fraud (inflated repair bills, exaggerated injuries) is subjectively labelled — inter-rater agreement between investigators is only 70–80%. Claim amounts follow a heavy-tailed distribution: most claims are ₹10k–50k but catastrophic claims of ₹1Cr+ dominate total payout. Log transformation is required. Two completely different ML pipelines must be built, tuned, and combined into a single risk score.

Dataset Information

Dataset Source	Kaggle Auto Insurance Claims | Kaggle Insurance Fraud Detection | Kaggle Health Insurance Cross-Sell Prediction
Access URL	https://www.kaggle.com/datasets/buntyshah/auto-insurance-claims-data | https://www.kaggle.com/datasets/mastmustu/insurance-claims-fraud-data
Features (Dimensions)	40 – 200 features (policyholder demographics, vehicle/property details, claim history, repair shop network, time-to-report)
Dataset Size	15,000 – 50,000 claims (Kaggle auto) | 1 million+ (large insurer proprietary datasets)
Data Type	Tabular (claims records with mix of categorical and numerical features)

ML Algorithms Used (Mapped to Your Syllabus)

- Logistic Regression — fraud probability baseline (interpretable for investigator explanation)
- Naive Bayes — probabilistic fraud classification using claim attributes
- Decision Tree — rule extraction for investigator screening criteria
- Random Forest — primary fraud classification ensemble
- Gradient Boosting — high-precision fraud scorer with tuned threshold
- Ridge & Lasso Regression — log-transformed claim severity prediction
- PCA — reduce correlated policyholder and vehicle features
- K-Means — repair shop / claimant risk profiling (detect fraud rings)

Evaluation Metrics

- F1-Score (fraud classification — balance precision and recall)
- Confusion Matrix (False Negatives = missed fraud = financial loss)
- MAE, RMSE on log-transformed claim amounts

- Combined Risk Score = fraud_probability × predicted_severity
- Lift Curve (SIU efficiency: how many investigations needed to catch 80% of fraud?)

Step-by-Step Implementation Roadmap

1. Download auto insurance claims dataset from Kaggle
2. Separate into two target variables: fraud_reported (binary) and total_claim_amount (continuous)
3. Feature engineering: claim-to-premium ratio, days-to-report delay (>30 days is a red flag), prior claims count in 3 years, repair shop network flag (is shop on approved panel?)
4. Pipeline 1 — Fraud Classification: Decision Tree → Random Forest → GBM; tune threshold for Recall > 0.80
5. Pipeline 2 — Severity Regression: log1p transform amounts; train Ridge and GBM regressors; evaluate RMSE on original scale
6. Combined Risk Score: fraud_probability × exp(log_predicted_severity)
7. Rank all claims by Risk Score: top 10% go to SIU automatically
8. Evaluate business impact: SIU hit rate (what % of top-10% claims are truly fraudulent?)
9. K-Means on repair shops: detect fraud rings (shops with high average claim amounts + high fraud rates)

Future Scope & Extensions

> Extend to image-based damage assessment fraud detection — analyse submitted repair photos using computer vision to validate damage severity vs claimed amount. Build a social network fraud ring detection system using graph analysis — identify repair shops, doctors, and lawyers who co-appear on multiple fraudulent claims. Implement NLP analysis of claim descriptions: extract inconsistency signals between initial report, police FIR (if applicable), and repair invoice. Apply survival analysis to model time-to-claim patterns — fraudulent claims exhibit different reporting time distributions. Integrate telematics data (black box GPS in vehicles) to validate accident location, speed, and braking patterns against claimed accident scenario.